Part I

In-line measurement for the control of food-processing operations

3

Principles of colour measurement for food

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3.1 Introduction

Vision, the most studied of all the human senses, has intrigued investigators for as long as man's speculations or accomplishments have been recorded. Modern studies into the mechanism of vision and human colour perception began in the seventeenth century with the recognition that the eye's lens must somehow project an image onto the back of the eye. Newton's classic experiments on the refraction of light led him to conclude that the rainbow did not possess colour; rather it was the spectrum's rays that produced the sensation (Wright 1967). The rationality of arranging colours into orderly systems based on Newton's seven rainbow colours has resulted in many attempts to create colour atlases with equal spacing of adjacent colours; for example, the widely used Munsell system and the newer Swedish natural colour space. The former is based on five hues and the latter on the six unique perceptions of black, white, red, green, yellow and blue (Hard 1970).

From the nineteenth-century experiments of Maxwell, Young and Helmholtz in mixing coloured lights (MacAdam 1970), it became clear that people with normal colour vision must have at least three retinal pigments in their eyes. By the late 1920s the eye's sensitivity to light relative to wavelength was established, the so-called 'standard observer' defined (Wright 1980) and the Commission Internationale de l'Éclairage (CIE) system of colour measurement constructed. Since then, considerable improvements have been incorporated into the system to make it nearly visually uniform. With the development of the computer, complex colour measurements and calculations are now routinely used for such industrial processes as paint formulation (Best 1987) and the colour match prediction (Nobbs 1997) and control of the appearance of dyed textiles (McLaren 1986; McDonald 1997). Instrumental colour measurement is increasingly being used for ingredient standardization and product quality control in the food industry.

The three interacting factors required to classify, estimate and measure the colour appearance of any object in a scene are an understanding of the human visual process, the effect of light on the environment and the nature of the material observed.

3.2 Colour vision: trichromatic detection

The sensation of colour vision can be thought of as the sum of the responses recognized in the brain from the signals detected by the eye of the scene viewed. The sensation is perceived as though it were projected out into the world from which it originated. This can lead to the error of imputing to the scene the sensations it generated. Sensations exist in the mind and not in the external world which produces them.

Human eyes, with their near circular field of view, are composed of three membranes (Fig. 3.1a). The outer membrane, the sclera, is continuous posteriorly with the sheath of the optic nerve and anteriorly with the cornea. The iris and the ciliary body, which suspends the lens, arise out of the middle layer, the choroid, which contains the capillary network. The inner membrane, the retina, lines the inside of the posterior of the eye. The first step in the visual process is the control of the amount of light entering the eye by the iris. The flux passes through the lens to be focused onto the fovea in the central region of the retina where it is detected. The signal is amplified (Normann and Werblin 1974) and

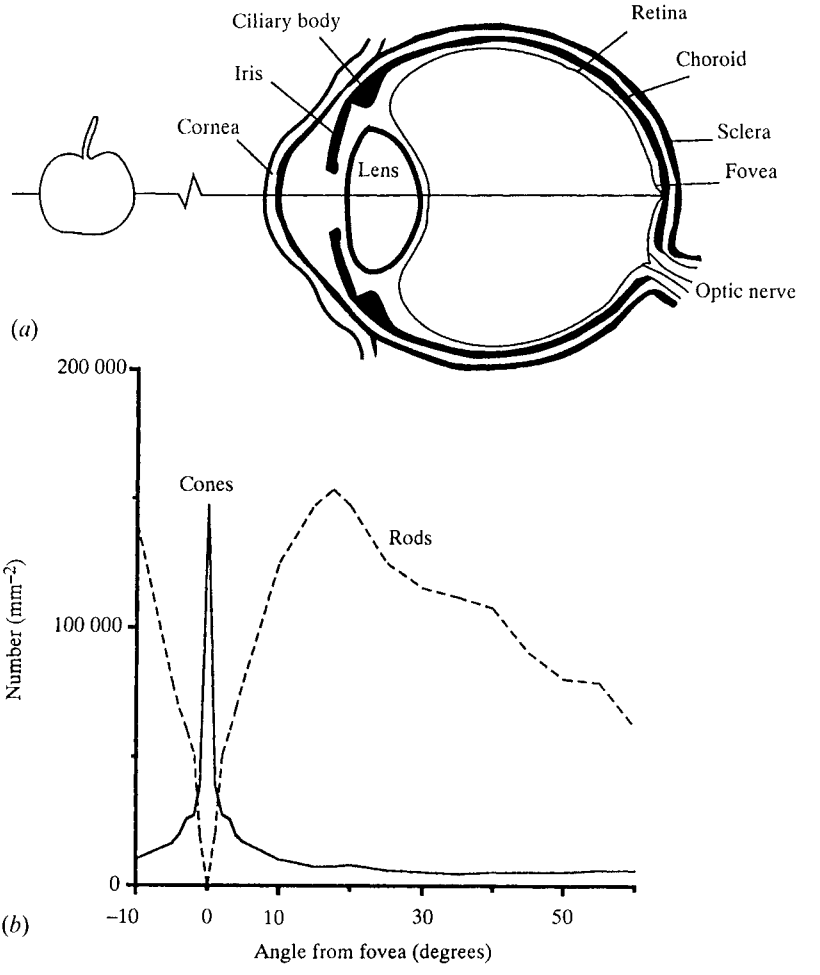


Fig. 3.1 Detection of light in the eye: (a) structure of the human eye (b) distribution of cones and rods in the temporal side of the retina (nasal side is similar except for the blind spot between 12° and 18° from the fovea).

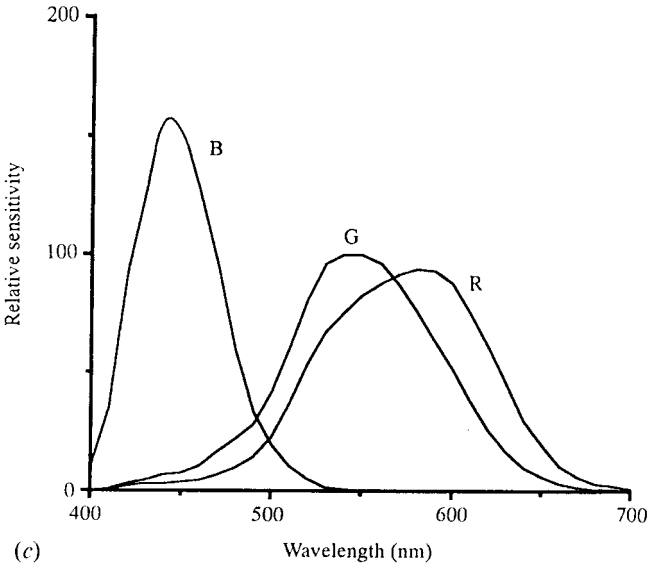


Fig. 3.1 continued. (c) spectral sensitivites of blue (B), green (G) and red (R) cone pigments.

then transmitted through the visual pathway (Rodieck 1979) for interpretation in specific areas of the visual cortex of the brain (Zeki 1980; Zeki 1993; Hubel 1988). The retina has two types of light-detecting receptors, the cones and the rods, so named because of their shape. The 'photoptic' colour-detecting cones, which are sensitive to three wavelength ranges, are densely packed in the centre of the fovea, which occupies < 2° of the visual field and is the basis of the so-called CIE 1931 2° standard colorimetric observer. The 'scotoptic' colourless detecting rods, which are over 100 times more sensitive to light than the cones, increase in density to 20° from the fovea and then decrease towards the periphery of vision (Fig. 3.1b). The 1964 10° supplementary standard observer was created to accommodate the improvement in colour discrimination as the visual angle increases beyond 2°; some rods are included in the detecting field.

Light energy, focused on the retina, is converted into electrical signals by conformational changes in the photopigments in the multifolded disk-shaped structures in the outer segments of the rods and cones (Wald 1968; Hurvich 1981; Jacobs 1981; Stryer 1988). Although only the rod pigment rhodopsin has been characterized, determination of the spectral absorption of the cone pigments has been possible using retinal densitometry with colour-blind observers deficient in one pigment (Smith and Pokorny 1975). The sensitivity curves of human cones determined by Estévez (1982) are presented in Fig. 3.1c. Rhodopsin absorbs maximally at 505 nm and the so-called blue (B), green (G) and red (R) cone pigments at approximately 440, 540 and 570 nm respectively. The sensitivity range of B marginally overlaps G and R between 450 and 550 nm, whereas the G and R functions substantially overlap, displaced from each other by only 20 to 30 nm. Thus monochromatic light at 580 nm which is near maximum for R appears yellow and not red because of the combined contributions of G and R. Increasing the wavelength to > 600 nm increases the contribution of R relative to G and the perceived colour becomes more red.

Cone vision is trichromatic; this means that any coloured light can be matched by a suitable mixture of red, green and blue primary lights, which is what would be expected from a three-receptor system. The actual colour-matching functions depend on the

wavelengths used as primaries. Although the initial detection of the stimulus is trichromatic, subsequent post-retinal processing gives rise to achromatic lightness/darkness and coloured red/green and blue/yellow opponent mechanisms (Hurvich 1981; Hunt 1995). The lightness component consists of a weighted summation of all three cone pigment absorptions, whereas it is the degree of differences among the B, G, R absorptions that generates the opponent colour mechanism. However, the neural linkages among the pigment cone signals are not in simple one-to-one opposition. The simplest scheme that can be constructed is that the red/green opponent response is red activated by absorption of B plus R and green activated by G; yellow is activated by G plus R in opposition to blue activated by B.

3.3 Influence of ambient light and food structure

3.3.1 Adaptation and colour constancy

Adaptation conditions the visual system to the chromatic nature of the surroundings as affected by the quality and intensity of the illumination. It compensates for changes in the spectral power distribution of the light and serves to keep the eye in balance (Boynton 1979). The magnitude of adjustment that chromatic adaptation has on visual experience is not usually realized because of the limitations of human memory for individual colours and the phenomenon of colour constancy (Brill and West 1986). White objects are recognized as white over a vast range of light intensity, from bright unshaded sunlight to the relatively dim levels of light found in room interiors, while coloured objects tend to have similar, but not necessarily identical, colours under most types of white illumination. Studies into the phenomena of adaptation which elicit this near constancy of colour appearance have been concerned mostly with predicting the changes that occur to colour recognition when lamp type and output are altered (Bartleson 1979a).

Lightness and contrast among greys are affected by luminance while colourfulness increases with the level of illumination and varies with the spectral emission of the lamp and its colour temperature (Hunt 1977). This is further discussed in Section 3.6.1 on fresh meat, where the degree of red enhancement of lamp spectra is shown to affect the perception of product attractiveness. Models of cone adaptation response have been used to predict the consequences of changing lamp spectra on object appearance (Bartleson 1979b; Nayatani *et al.* 1986; Hunt 1987a). The concept of colourfulness has been used to construct grids of constant hue from which other grids can be derived for other illuminants (Pointer 1980, 1982). Such models use logarithmic and hyperbolic functions to mimic the physiological mechanisms involved. Hunt's (1987a) model can be used to predict the changes that occur in object colours at any level of illumination for a wide range of backgrounds in the realistic situation where the eye's fixation wanders. These effects of light quality on colour perception illustrate the difficulties in separating the concept of vision from that of appearance. Vision is modulated by the light from the scene, while the characteristics of appearance are modified by the light incident upon the object. Hence systems for classifying appearance and procedures devised for both observing and measuring colour must take account of the nature, quality and quantity of the light as it affects the observer's perception. The British Standards Institute and the International Standards Organisation have recently produced general guidance and test methods of the assessment of the colour of foods (BSI 1999; ISO 1999).

3.3.2 Appearance

Colour is usually considered the most important attribute of any food's appearance (Francis and Clydesdale 1975) especially if it is associated with other aspects of food quality, for example the ripening of fruit or the visible deterioration which occurs when a food spoils. Nearly every food product has an acceptable range of colour, which depends on a wide range of factors which include variability among consumers, their age and ethnic origin, and the physical nature of the surroundings at time of judgement (Francis 1999). However, in addition to colour specification, the nature of internally scattered light and the distribution of surface reflectance are required for a more complete description of appearance. The food's structure and pigmentation interact to affect both translucency/opacity and colour; for example, small changes in scatter may produce larger changes in colour than are attributable to change in pigment concentration (MacDougall 1982).

The characterization of an object's appearance is accomplished in two stages. The first is physical and the second is psychological. The physical characteristics are the size, shape and uniformity of the object along with the type and degree of pigmentation and the nature of the object's structure that attenuates light. The physical information is converted to the psychological by translating the object's reflectance or transmittance spectrum into the tristimulus values and then to a defined colour space.

The concept of 'total appearance' can be applied to foods (Hutchings 1994). Total appearance is seen to comprise not only the physical appearance characteristics of the food but also includes such social factors as the observer's culture, memory, preferences and the human appreciation of the product. Foods have an infinite variety of appearance characteristics. Their surfaces may be diffuse, glossy, irregular or flat. They may be transparent, hazy, translucent or opaque and their colour may be uniform or patchy. Hence colour-measuring procedures for foods often have to be modified from those used in the measurement of flat opaque surfaces such as paint and paper for which most colour-measuring instruments are designed. Different instrument optical geometries will lead to difficulties in sample presentation, producing different colour values for the same material. The inclusion or exclusion of surface specular reflection in the measurement depends not only on its importance as a characteristic of the food but also on the design of the detector-sensor unit in the instrument. Lateral transmittance of light through translucent materials affects their reflectance (Atkins and Billmeyer 1966; Hunter and Harold 1988; MacDougall 1988; Hutchings, 1994) and must be allowed for in the assessment of such products as tomato paste (Brimelow 1987) because the ratio of absorption to scatter varies with aperture area and the concentration of components in the product (Best 1987; MacDougall 1987).

3.3.3 Absorption and scatter

The reflection of light from opaque and translucent objects depends on the ratio of absorption to scatter as affected by pigmentation, refractive index and the light-scattering properties of the material. The Kubelka-Munk method of separating subsurface absorption and scatter (Kubelka 1948) is fully illustrated by Judd and Wyszecki (1975). It relates reflectivity R_∞, i.e. reflectance at infinite thickness, to the coefficients of absorption K and scatter S by

$$K/S = (1 - R_\infty)^2/2R_\infty$$

which can be calculated from the reflectance of thin layers on white and black backgrounds. If K and S are required for prediction purposes, the accuracy of their measurement can be improved by appropriate correction factors for surface reflection (Saunderson 1942). Colour calculated from R_∞, with separate estimation of the specular component as gloss is usually sufficient information to describe opaque objects, but for translucent materials K and S are also necessary.

3.4 Colour description

3.4.1 The CIE system

The CIE system of colour measurement (ASTM 2000; CIE 1986) transforms the reflection or transmission spectrum of the object into three-dimensional space using the spectral power distribution of the illuminant and the colour-matching functions of the standard observer (CIE 1986). The mathematical procedure is given in any standard text on colour, for example Wright (1980); Judd and Wyszecki (1975); Hunt (1995). The system is based on the trichromatic principle but, instead of using 'real' red, green and blue primaries with their necessity for negative matching, it uses 'imaginary' positive primaries X, Y, and Z. Primary Y, known as luminous reflectance or transmittance, contains the entire lightness stimulus. Every colour can be located uniquely in the 1931 CIE colour space by Y and its chromaticity coordinates $x = X/(X + Y + Z)$ and $y = Y(X + Y + Z)$, provided the illuminant and the observer are defined.

The original illuminant representative of daylight was defined by the CIE as source C, but it has now been superseded by D65, i.e. an illuminant which includes an ultraviolet component and has a colour temperature of 6500 K. The colour temperatures of lamps and daylight range from approximately 3000 K for tungsten filament lamps and 4000 K for warm white fluorescent to 5500 K for sunlight and 6500 K for average daylight. Because the original 2° colour-matching functions apply strictly only to small objects, i.e.

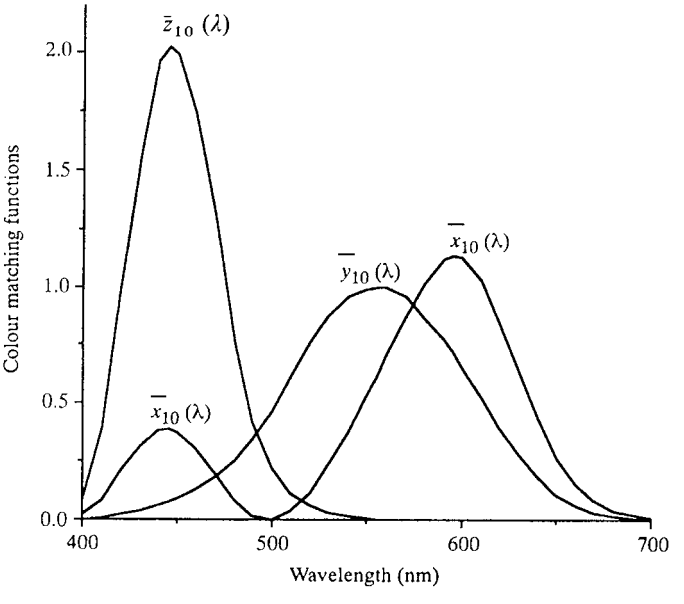


Fig. 3.2 Colour matching functions of the CIE 10° standard observer.

equivalent to a 15 mm diameter circle viewed at a distance of 45 cm, the CIE has added a 10° observer (Fig. 3.2) where the object diameter is increased to 75 mm. The current trend in colour measurement is to use D65 and the 10° observer. The 1986 CIE recommended procedures for colorimetry are included in the ASTM Standards (2000) and also in Hunt (1995) along with the weighting factors for several practical illuminants (Rigg 1987). These include representative fluorescent lamps, of which F2 is a typical lamp at 4230 K but with a low colour rendering index of 64 (Fig. 3.3). The colour rendering index R_a is a measure of the efficiency of a lamp at a given colour temperature to render the true

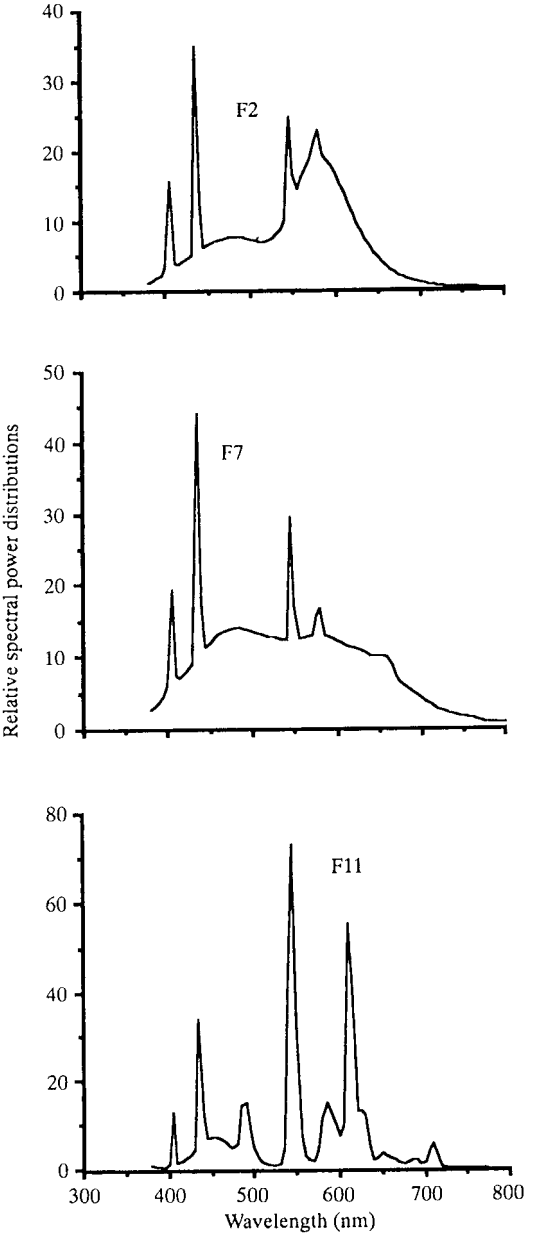


Fig. 3.3 Relative spectral power distributions of preferred CIE representative fluorescent lamps.

appearance of Munsell colours. The broadband lamp F7 has the same colour temperature (6500 K) and chromaticity coordinates as D65 and, because of its flatter spectrum, it has a high R_a of 90. The triband lamp F11 (4000 K) also has a moderately high R_a of 80, but its main advantage is its much improved efficiency in energy utilization.

3.4.2 Uniform colour space

The original 1931 CIE *Y*, *x*, *y* system of colour measurement is not visually uniform (Fig. 3.4a). Constant hue and chroma are distorted and equal visual distances increase several-fold from purple-red to green. Improved spacing has been accomplished by both linear and non-linear transformations of *Y*, *x*, *y* (Billmeyer and Saltzman 1981). Near uniform colour spaces of practical importance are the Hunter and the CIELUV and CIELAB spaces. In the Hunter (1958) *L*, *a*, *b* colour space the lightness coordinate *L* is the square

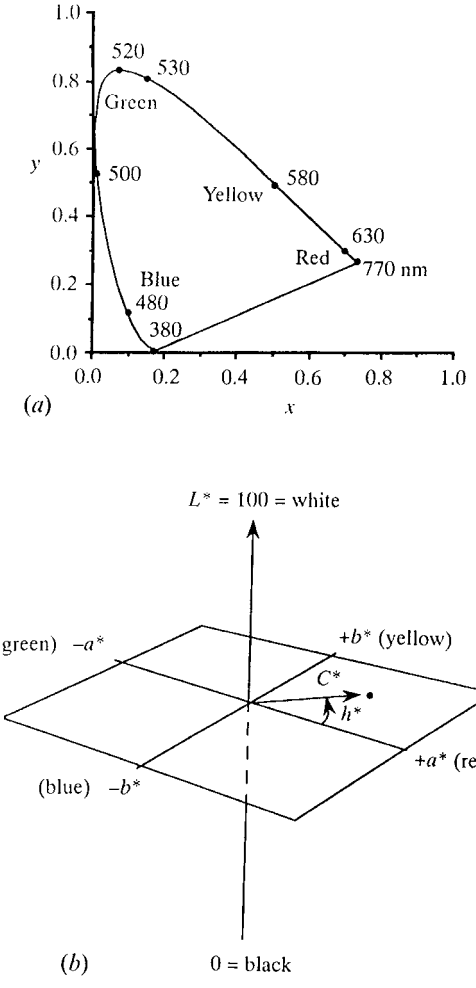


Fig. 3.4 Colour diagrams: (a) CIE 1931 chromaticity diagram showing non-uniformity of spacing of red, yellow and blue unique hues (b) CIELAB uniform diagram showing relationship of red/green (a^*+/−) and yellow/blue (b^*+/−) opponent coordinates to lightness L^*, chroma C^* and hue angle h^*.

root of the tristimulus value Y, and a, and b are the red/green and yellow/blue opponent coordinates. In the 1976 CIELUV and CIELAB spaces (Robertson 1977), which were an attempt to reduce the many scales then in use to two, the lightness coordinate L^* is the same for both but the spaces use different concepts in their construction. The CIE L^*, a^*, b^* space (Fig. 3.4b), known as CIELAB, has generally replaced the Hunter space for industrial applications. The improvement is due to the non-linear cube root transformation of the 1931 tristimulus values which approximates the visual spacing of the coloured samples in the Munsell system. The formulae are

$$L^* = 116(Y/Y_n)^{1/3} - 16 \quad \text{for } Y/Y_n > 0.008856$$
$$L^* = 903.3(Y/Y_n)^{1/3} \quad \text{for } Y/Y_n < 0.008856$$
$$a^* = 500[(X/X_n)^{1/3} - (Y/Y_n)^{1/3}]$$
$$b^* = 200[(X/X_n)^{1/3} - (Z/Z_n)^{1/3}]$$

where X_n, Y_n, Z_n refer to the nominally white object colour stimulus.

Newer scales for determining small differences (BSI 1988), as used in the textile industry, are now included in the software of some automatic colour-measuring spectrophotometers. For example, the CMC(l:c) colour difference equation, where weighting factors are applied to the CIELAB $L^*C^*h^*$ data, produces a more uniform colour acceptability scale.

3.4.3 Further terminology

Colour terms can be divided into the subjective and the objective (Hunt 1978). The subjective, the psychosensorial, are brightness, lightness, hue, saturation, chroma and colourfulness. Colourfulness, a recently introduced term, is that aspect of visual sensation according to which an area appears to exhibit more or less chromatic colour. Although hue is easily understood as that attribute described by colour names – red, green, purple, etc. – the difference between saturation and chroma is less easily comprehended. Saturation is colourfulness judged in proportion to its brightness, whereas chroma is colourfulness relative to the brightness of its surroundings. A similar difference exists between lightness and brightness. Lightness is relative brightness. Lightness is unaffected by illumination level because it is the proportion of the light reflected, whereas the sensation of brightness increases with increase in illumination.

The objective terms, the psychophysical, refer to the stimulus and are evaluated from spectral power distributions, the reflectance or transmittance of the object and observer response. They provide the basis for the psychometric qualities which correspond more nearly to those perceived. For CIELAB space the more important terms are lightness L^*, hue $h^* = \tan^1 (b^*/a^*)$ and chroma $C^* = (a^{*2} + b^{*2})^{1/2}$. Total colour differences ΔE^* can be expressed either as the coordinates of colour space or as the correlates of lightness, chroma and hue. Hence

$$\Delta E^* = [(\Delta L^*)^2 + (\Delta a^*)^2 + (\Delta b^*)^2]^{1/2}$$

or

$$\Delta E^* = [(\Delta L^*)^2 + (\Delta C^*)^2 + (\Delta H^*)^2]^{1/2}$$

where ΔH^* is used rather than Δh^* because the latter is angular. For small colour differences away from the L^* axis, if h^* is expressed in degrees, then

$$\Delta H^* = C^* \Delta h^* (\pi/180)$$

Table 3.1 Overview on colour description systems and notation

CIE system (1931)
This is based on the imaginary positive primaries *X*, *Y*, *Z* (transformed from real red, green and blue functions of trichomatic stimulus detection which may contain negative values).
In CIE space, colour is located (assuming defined illuminant and observer) by (*Y*, *x*, *y*), where

Y	luminous reflectance or transmittance (containing the entire lightness stimulus)
x, y	chromaticity coordinates

$$x = X/(X + Y + Z)$$
$$y = Y/(X + Y + Z)$$

CIE space is not visually uniform.

Hunter Lab System (1958)
In Lab space, colour is defined by (*L*, *a*, *b*), where

L	correlate of lightness
a, b	red/green and yellow/blue opponent coordinate correlates

$$L = 10\ Y^{1/2}$$
$$a = [17.5(1.02X - Y)]/Y^{1/2}$$
$$b = [7.0(Y - 0.847\,Z)]/Y^{1/2}$$

CIELAB system (1976)
In CIELAB space, colour is defined by (*L**, *a**, *b**), where

L^*	visually uniform lightness
a^*, b^*	visually uniform chromaticness coordinates

$$L^* = 116(Y/Y_n)^{1/3} - 16 \quad \text{for } Y/Y_n > 0.008856$$
$$L^* = 903.3(Y/Y_n) \quad \text{for } Y/Y_n < 0.008856$$
$$a^* = 500[(X/X)_n^{1/3} - (Y/Y_n)^{1/3}]$$
$$b^* = 200[(Y/Y_n)^{1/3} - (Z/Z_n)^{1/3}]$$

Where X_n, Y_n, Z_n are the values of X, Y, Z for the reference white.

Further terms used are

$$h^* = \tan^{-1}(b^*/a^*) \quad \text{hue}$$
$$C^* = (a^{*2} + b^{*2})^{1/2} \quad \text{chroma}$$

Other terms, such as adaptation, appearance, vision colourfulness, saturation and brightness, are discussed in Sections 3.3.1, 3.3.2 and 3.4.3.

A comparison of the major colour scales with the associated terminology is given in Table 3.1.

3.5 Instrumentation

Since colour is a psychological phenomenon, its measurement is based on human colour perception. Hence, photoelectric instruments must be corrected for both lighting and human visual response, while visual techniques must use observers with 'normal' colour vision under defined lighting. Examples of direct visual assessment are colour atlases for broad definition of the location of colours in colour space, collections or sets of printed or painted coloured papers specific to products or processes, and visual matching instruments

which use coloured filters. Typical of the former are the Munsell and Swedish NCS atlases which are structured on uniform colour space, and the Pantone collections of printer's colours with defined ink mixtures printed from 10 to 100 per cent tinting strength. Probably the best known of the visual matching instruments is the Lovibond tintometer in which the object, under specified illumination, is viewed and matched against a series of coloured filters interposed over a white background by the observer.

Photoelectric colour-measuring instruments can be divided into two classes, trichromatic colorimeters and spectrophotometers. The most successful of the early trichromatic colorimeters was developed in the 1940s by Hunter (1958). It comprised a light source and three wideband red, green and blue filters to approximate CIE standard illuminant C and the 2º observer. The tristimulus values obtained were transformed into Hunter *L*, *a*, *b* colour space. Until the advent of the computer and the photodiode such instruments were much less expensive than spectrophotometers and, although absolute accuracy may have been poor, they were very good at measuring the small colour differences demanded for industrial process control (Patterson 1987).

The more modern tristimulus instruments are linked to computers with automatic calibration and the provision of a number of colour spaces. Such instruments may be supplied with a selection of sensor heads of different illuminating geometries to allow measurement of a wide range of product types depending on the nature and dimensions of their surfaces. A range of hand-held lightweight colorimeters and miniature diode array spectrophotometers, with optical geometries comparable in function with the larger bench instruments, is now manufactured by several companies. Their compactness is a direct result of the use of high energy pulsed xenon arc lamps combined with filtered silicon detectors and microchip circuitry. The comparative inexpensiveness of such instruments, with their built-in memories and choice of colour scales, has resulted in increased in-line colour measurement in all branches of industry where colour control is necessary or desirable.

The most accurate colour-measuring instrument is the spectrophotometer. Reflectance instruments are usually fitted with an integrating sphere with the choice of including or excluding the specular component of reflectance. Care must be exercised in deciding which geometry is appropriate for particular applications. The diffuse component of reflectance from subsurface absorption and scatter is wavelength dependent, whereas the specular component is not. For materials with glossy surfaces the inclusion of the specular will increase measured reflectance which, when translated into colour space, can lead to large discrepancies in the interpretation of visual lightness, as usually viewed, and to a lesser extent of the chromaticness of the colour. For example, highly glossy black tiles used for instrument calibration have tristimulus *Y* values of approximately 0.3 when the specular is excluded but 4.5 when included. The consequence of this difference in *Y* of 4 per cent produces a specular excluded uniform lightness *L** of 3 and an included *L** of > 25. For medium grey and white tiles the excluded to included *Y* values are approximately 25 to 29 and 78 to 82 repectively, which give *L** values of approximately 57 to 61 and 91 to 92 respectively. Hence the near constant effect of 4 per cent on *Y* from the specular reflectance produces a decreasing effect from black to white from > 20 to about 1 per cent in *L**. The CIE recommends that colorimetric specifications of opaque materials should be obtained with one of the following conditions of illumination viewing geometries and should be specified in any report:

- 45º/0 or 0º/45º, specular excluded
- diffuse/0º or 0º/diffuse, specular included or excluded.

However, the spectrophotometers most commonly used for measuring colour do not have identical geometries. Three typical instruments have been compared by Patterson (1987), who points out that probably the biggest source of differences among the instruments can be traced to the specular component. Hunt (1987b) suggests that if measurements are to be compared it is better to include the specular because of the considerable variation in the area of gloss traps used in different spheres, but the more nearly correct measurements in relation to practical visual observation is with the specular excluded (Best 1987). For computer match prediction of pigmented materials, e.g. paint formulation, the total reflection (i.e. specular included) is preferred. This restriction does not usually apply to tristimulus colorimeters which normally exclude the specular component of reflectance where the illumination viewing geometry is 45º/0º, as in the classic Hunter bench colorimeter.

Another important source of variation among tristimulus colorimeters and spectrophotometers is the area of the viewing aperture and the relative area of the illuminating light spot, which affects both the direction and the amount of light returned from translucent materials. MacDougall (1987) has demonstrated that translucent suspensions of milk exhibit a ten-fold decrease in K/S for an increase in aperture area from 5 to 20mm. Best (1987) states that accurate determination of K and S by measuring thin layers on black and white backgrounds requires that the ratio of the aperture area to the thickness of the sample must be considerably greater than 10, a criterion unlikely to be met for most foods. One further source of potential error, in addition to those associated with instrument geometry and sample structure, is the wavelength interval used to calculate the tristimulus values. Although the CIE (1986) specifies the standard observer at 5 nm intervals from 380 to 780 nm, such accuracy is not required for most practical purposes. For 10 nm accuracy the intermediate 10 nm values from the 5 nn tables should be used. However, the CIE has not yet officially recommended the use of 20 nm intervals, although many modern colour spectrophotometers detect at 20 nm intervals. Tables of weighting functions at 20 nm intervals for both the CIE illuminants and a variety of fluorescent lights have been calculated and are published in the up-to-date colour textbooks cited in this chapter. Errors attributable to wavelength interval are likely to be less important than those from instrument geometry, except when estimating the effects of narrowband emission lamps on materials with several absorption bands. Here the 20 nm interval may prove to be less effiecient.

3.6 Examples

The progress of pigment oxidation in fresh meat and the effects of illumination on orange juice are given as examples of the interaction of absorption and scatter on measured colour and visual appearance. The effect of the illuminant on the calculation of CIELAB from a variety of food specta is presented.

3.6.1 Fresh meat

On exposure to air the purple ferrous haem pigment myoglobin on the surface of freshly cut meat oxygenates to the bright red covalent complex oxymyoglobin. During refrigerated display, oxymyoglobin oxidizes to brownish green metmyoglobin (MacDougall 1982; MacDougall and Powell 1997). Twenty per cent of surface oxymyoglobin with metmyoglobin causes the product to be rejected at retail because of its faded colour

(Hood and Riordan 1973). The changes in the mean reflectance spectra of over 100 packages of beef overwrapped with oxygen permeable film and held in the light at < 5C over a period of 1 week are shown in Fig. 3.5. As the pigment oxidizes there is an increase in reflectance in the green region of the spectrum as the alpha and beta absorption bands decrease and a distinct loss in reflectance in the red region with development of the metmyoglobin absorption band at 630 nm. The changes in colour, calculated for CIELAB and D65, are shown in Fig. 3.6. As meat fades there is a small loss in lightness L^*, accompanied by much greater changes in the chromaticness coordinates a^* and b^*. The loss in a^* and gain in b^* can be interpreted as an increase in the hue angle h^* in the direction of yellow; however, since it occurs with a concomitant loss in chroma C^*, it is recognized as a more grey or dull colour. Dull yellow is perceived as brown.

The appearance of meat is greatly affected by the colour rendering properties of the lamps used for display (Halstead 1978). Some fluorescent lamps recommended by the lamp industry for displaying meat have enhanced red emission which tend to maintain the preferred colour of oxymyoglobin and visually shifts the early stages of metmyoglobin development from brown towards red. This effect of red enhancement on meat colours has been shown to elicit a greater visual colour change in making brown appear red than in making red appear more red (MacDougall and Moncrieff 1988).

For some people the flattering of red-enhanced lamps may make meat appear too red. The ICS Micro Match spectrophotometer used to measure these samples is equipped with the option of using alternative illuminants to calculate CIELAB. The estimated changes in meat colour attributable to different illuminants after 1 and 4 days' exposure (Table 3.2) illustrate the effect that light quality has on lightness, hue and chroma. The changes in colour produced by the differences in colour rendering among some of the lamps are

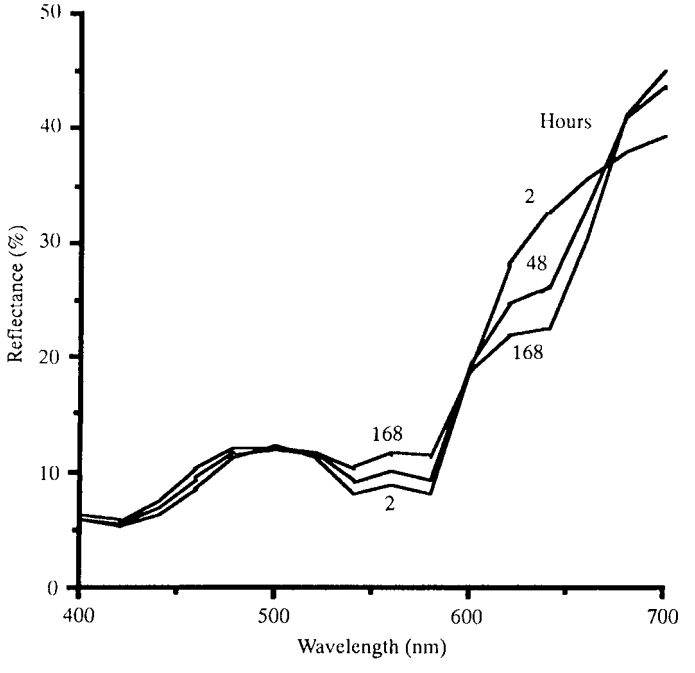


Fig. 3.5 Reflectance spectra of fresh beef during oxidation of oxymyoglobin to metmyoglobin obtained on a diode array spectrophotometer at 20 nm intervals: means of over 100 samples wrapped in oxygen permeable film and stored at <5ºC under 1000 lux fluorescent illumination for 1 week.

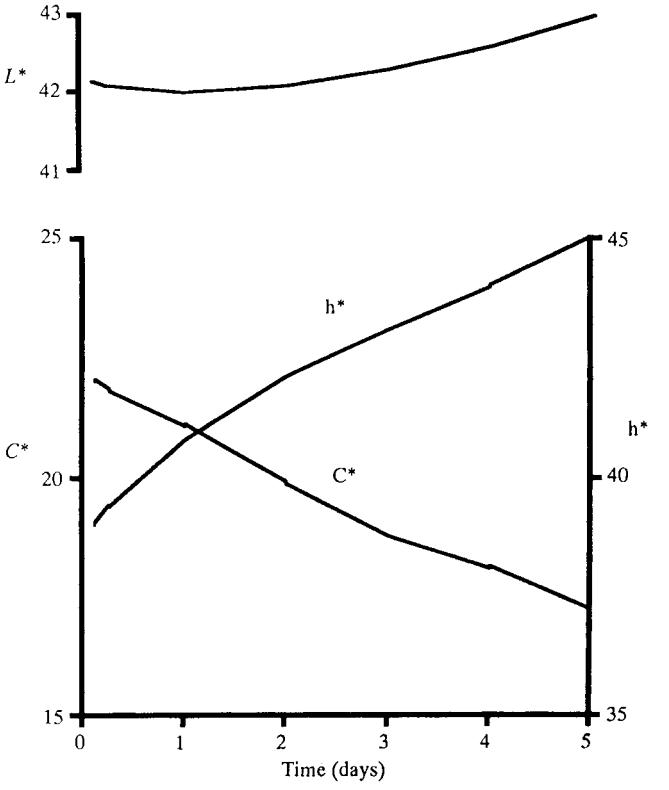


Fig. 3.6 Progressive changes in lightness L^*, hue angle h^* and chroma C^* calculated from spectra of wrapped fresh beef stored at <5°C under 1000 lux fluorescent illumination during oxidation of surface oxymyoglobin to metmyoglobin.

equivalent to that which occurs after 4 days' fading, that is $\Delta L^* \leq 1$, $\Delta C^* \leq -4$ and $\Delta h^* \leq 7$. There was little change in L^* on changing illuminant, but the large changes in C^* and h^* illustrate the effects of decreasing lamp colour temperature, altering flattery and improving fidelity. A decrease in colour temperature from D65, as red emission increases, generally increases C^*, that is the colour is perceived as brighter or more

Table 3.2 Calculated changes in L^*, C^* and h^* from D65 to other lamps for the fresh beef spectra shown in Fig. 3.5

Storage time		Difference in colour from D65					
		ΔA	ΔWWF	ΔNFL	ΔCWF	Δ83	Δ84
2 hours	ΔL^*	2.9	2.0	0.6	2.0	2.2	1.2
	ΔC^*	7.3	7.1	2.2	7.1	7.6	4.9
	Δh^*	−2.3	−1.2	3.9	6.5	3.3	−2.3
4 days	ΔL^*	2.3	1.6	0.6	−0.2	1.7	0.9
	ΔC^*	5.7	5.5	2.1	−0.7	6.3	3.8
	Δh^*	−5.9	−1.9	4.5	9.1	2.8	−3.5

Tungsten lamp: A. Fluorescent lamps: WWF, warm white; NFL, natural; CWF, cool white; 83, triband at 3000 K; 84, triband at 4000 K.

intense with the observer adapted to white, but h^* may become more brown (more positive) or more red (more negative) as influenced by both colour temperature and the lamp's spectral bandwidth which affects fidelity.

3.6.2 Orange juice

Translucent suspensions are difficult to measure, and direct unobserved interpretation of instrumental data can lead to confusion because of the way the incident light is dispersed in the sample. Most consistent results are obtained if the instrument aperture is large relative to the incident beam (Kent 1987; MacDougall 1987). The effects of optical geometry on colour and the Kubelka-Munk absorption K (mm^{-1}) and scatter S (mm^{-1}) coefficients for orange juice were studied by MacDougall (1983), who found Y_∞, the luminous reflectance at calculated infinite thickness, increased by 50 per cent if the aperture diameter was increased from 2 cm to 5 cm while the incident beam was maintained at 1 cm.

The effect of dilution of four-fold orange juice concentrate on the reflectance spectra obtained on 4-cm thick samples in thin-walled polystyrene bottles is shown in Fig. 3.7. As can be seen, 4 cm is practically equivalent to infinite thickness. Kubelka-Munk absorption and scatter coefficients were calculated from 2-mm thick samples on black and white backgrounds. On dilution, S decreased for X, Y and Z as the suspension became more translucent. K for Z decreased to approach the much lower near constant values of K for X and Y (Fig. 3.8). This decrease in K for Z is as anticipated for a blue absorbing pigment. The effect of loss of scattering power on dilution was to reduce Y_∞, and hence lower L^*. The most dilute juice, therefore, is instrumentally the darkest, and the most concentrated is the lightest (Fig. 3.9). However, this is not what is perceived. Glasses of orange juice viewed with overhead illumination range from pale yellow for concentrations less than 1 to deep orange at a concentration of 4, which is opposite to that determined instrumentally. For strongly scattering coloured materials in dilute

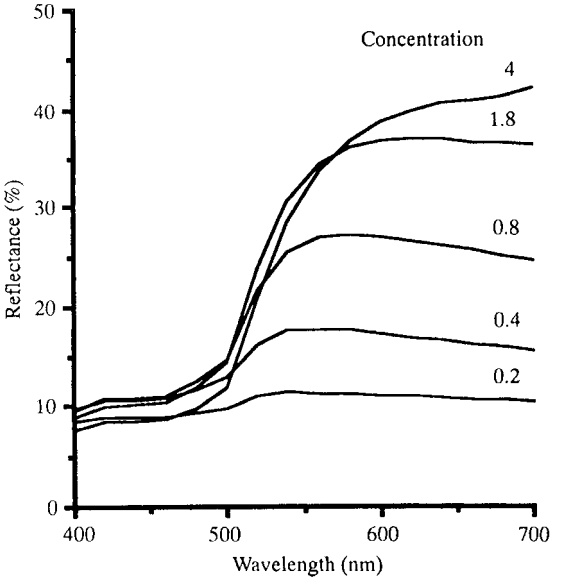


Fig. 3.7 Reflectance spectra of concentration and diluted orange juice at a path length of 4 cm, equal to infinite thickness: reconstituted juice at normal concentration = 1.0.

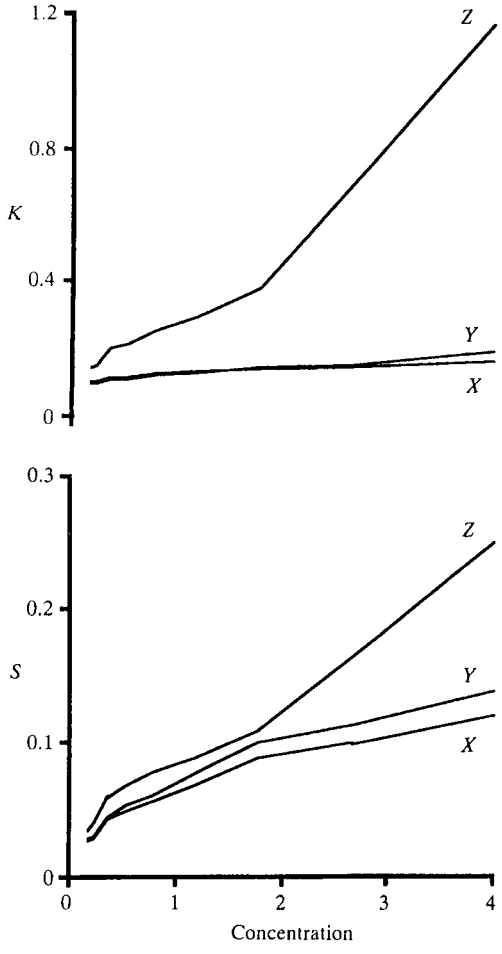


Fig. 3.8 Kubelka-Munk absorption *K* and scatter *S* coefficients for tristimulus values *X*, *Y* and *Z* for concentrated and diluted orange juice: values (mm^{-1}) calculated from reflectance spectra obtained from 2 mm path length cells with black and white backgrounds.

suspension, measured colour, even supplemented by information on scatter, is inadequate fully to describe appearance. The instrument does not measure what the observer sees because light is reflected from a limited solid angle, whereas the observer's perception is influenced by the multidirectionality of illumination which makes coloured translucent materials appear to glow.

3.6.3 Illuminant spectra and uniform colour

Although D65 is the illuminant spectrum most used for calculating CIELAB, of the other lamps listed in Table 3.2 the more important for relating food colour measurement to visual colour in practice are A and 83. A is the emission spectrum of tungsten illumination and 83 is that of 'tri-band' fluorescent illumination, similar to F11 in Fig. 3.3. However in 83 the red phosphor is increased by >25% over that of F11 and the green phosphor reduced by about 12% giving the lamp a warm appearance

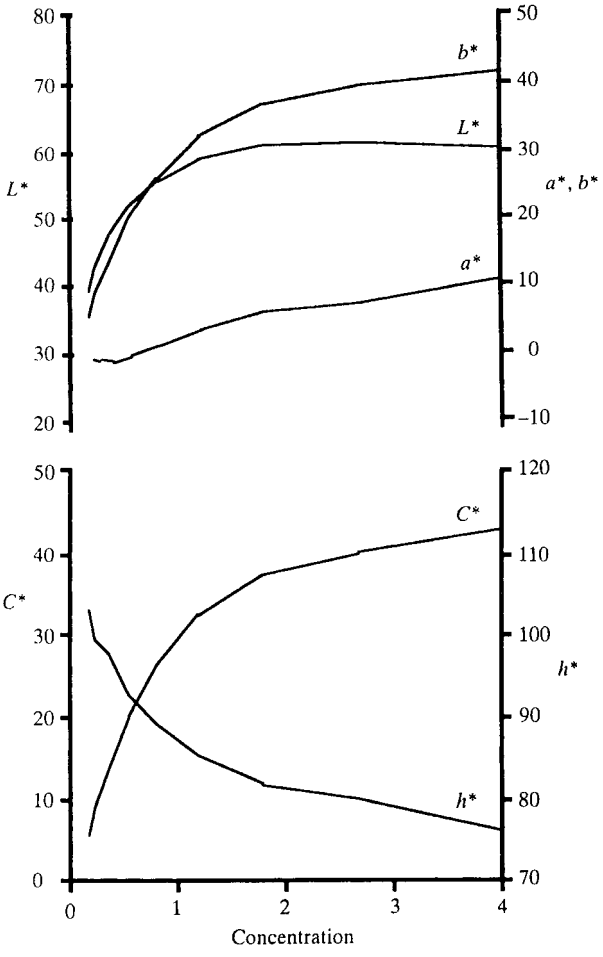


Fig. 3.9 Changes in lightness L^*, opponent coordinates a^* and b^*, hue angle h^* and chroma C^* calculated from reflectance spectra of concentrated and diluted orange juice at 4 cm path length.

(3000°K) similar to that of tungsten but with a considerable reduction in energy. 83 is now commonly used for food display in supermarkets because of its attractiveness and low heat output.

To examine the size of the effect of these lamps on CIELAB colour space, a variety of foods, selected as representative of the major hues, were measured on the ICS Micro Match spectrophotometer (Fig. 3.10). The red hue examples were fresh tomatoes at supermarket readiness, i.e. not excessively ripe but bright attractive red-pink and Wiltshire bacon, typically translucent in appearance. The yellow spectrum was the mean of several bananas, ripe but without any indication of brown spotting. The difference between the two orange samples was that the colour of the peel of entire navel oranges is near opaque and brilliant, whereas the colour of freshly squeezed orange juice is translucent. This results in a large reduction in the juice's reflectance at the red end of the spectrum compared to that of the peel. The green spectrum was that of the external leaves of a mid to light green cabbage and the brown was that of the surface of a 'digestive' biscuit. The near-white spectrum was that of semi-skimmed milk. All samples were measured at infinite thickness.

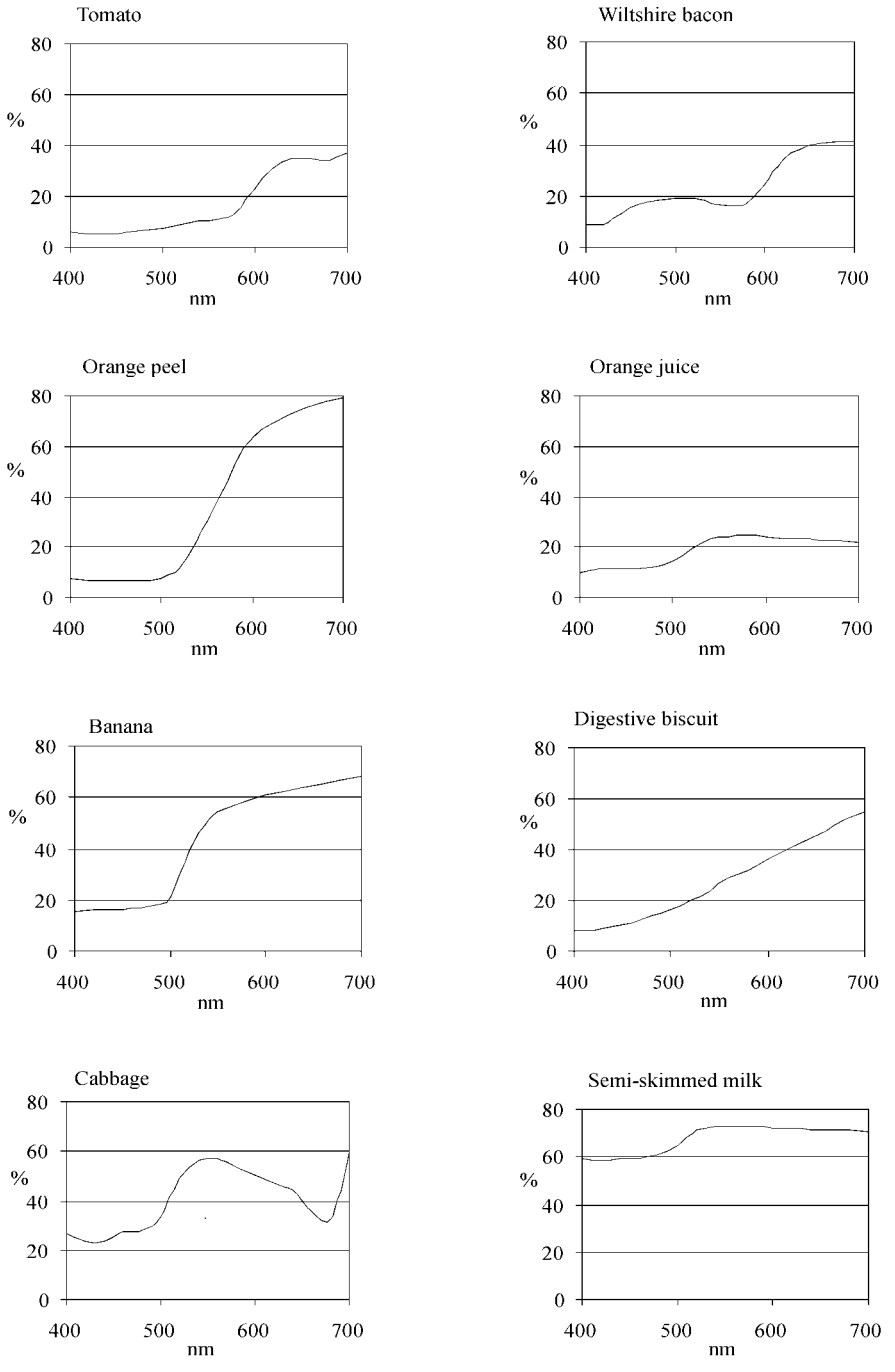


Fig. 3.10 Reflectance (%) spectra of a selection of foods of differing hue.

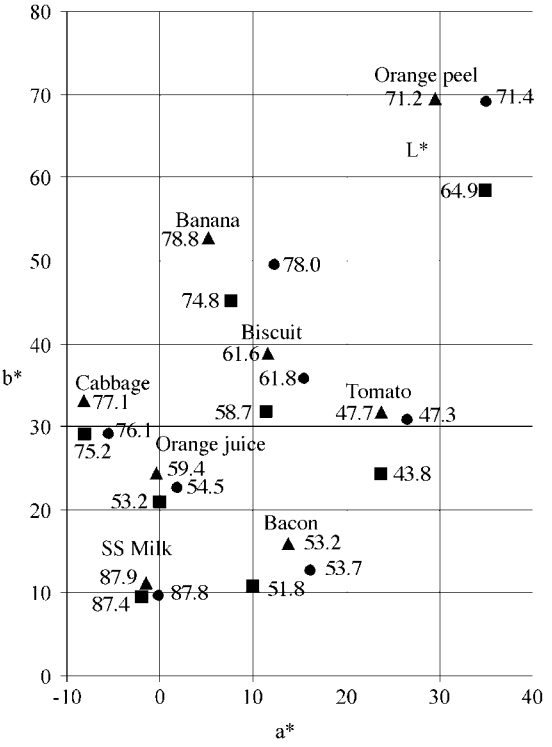


Fig. 3.11 CIELAB a^*b^* spacing of the food spectra from Fig. 3.10 calculated for the illuminants D65 (■), A (●) and 83 (▲). The L* values for each food are given adjacent to the symbols of the illuminants.

Figure 3.10 shows the average reflectance spectra of the foods and Fig. 3.11 the location of the samples in CIELAB calculated for the three illuminants. The triplets of points for each sample show that D65 has lower b^* values, that is more blue as would be expected, and A and 83 are more yellow with A more red than 83 with higher a^* values. These results parallel visual observation of the products viewed under tungsten, artificial daylight (D65) and Philips 83 lamps. This study clearly demonstrates that if interpretation of CIELAB is to be related to illumination conditions in practice, then D56 should not be the only illuminant used to calculate food colour.

3.7 Conclusions

This chapter has attempted to present the basics of colour measurement as applied to foods. It is important to realise that the wide variability in the nature of foods and food products, from both their structure and pigmentation, may limit any colour-measurement technique applicable only to that particular food. This is because the numeracy of the data is unlikely to match the visual experience of equivalent reference atlas colours. In some cases this discrepancy may be large and has to be recognised as an intrinsic property of the food. These differences may arise from the difference of the visual experience of the product when viewed under normal lighting conditions as opposed to the limitations of its optical properties when presentated to the particular

colour-measuring instrument. This is particularly so in the case of translucent foods. The values of measured lightness, hue and chroma are likely to be quite different from similarly coloured opaque materials. A consequence of this is that, if required for quality decisions, the variability of measured colour for any food over the range of acceptability and unacceptability needs to established relative to real samples judged under appropriate controlled visual judgement conditions. If the foregoing are taken account of, the use of colorimetry is likely to expand in the food industry in the future, especially so as portable instruments, and in-line measurements techniques, become accepted as reliable indicators of product quality.

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